# AGFE-Fire v4: Adaptive Group-Weighted Fusion Ensemble for Forest Fire Vulnerability Prediction

Our Proposed Methodology — Version 4 (Final, Prof. Kiran's Guidance)

Ensembling ML Models to Identify Forests Vulnerable to Fire
Across Selected Biogeographic Zones of India (2003–2025)

Document 5 v4 — Every method tagged with source

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**Professor Kiran's Key Directives — Incorporated in This Version**

1. **Study units = Biogeographic zones** (from India's 16 major forest types, coupled with biogeography). Select a **few** zones as fundamental units for model calibration/validation.
2. **Temporal window = 2003–2025** (22 years). Consistent MODIS fire data from 2003. Needed to capture climate influence over 2–3 decades.
3. **ENSO cycle (4–5 years)** influences rainfall and drought — must be captured in temporal modeling.
4. **80/20 split** for calibration/validation.
5. **Increase temporal, decrease spatial** — deeper analysis on fewer zones rather than shallow analysis on all India.
6. **Data volume:** Full India could be ∼500K fire points — computationally intensive. Subset of zones keeps it manageable for a semester project.

**Source Tags**

- **[Doc1]** = Sarkar et al. (AUC-Weighted Ensemble, India, NEI)
- **[Doc2]** = Shahzad et al. (Multi-Layer Stacking + SHAP)
- **[Doc3]** = BIPE — Hu et al. (Bi-Layer Ensemble, Ensemble > DNN)
- **[Doc4]** = FireRisk-Multi — Yuan et al. (Dynamic Multimodal Fusion, GEE)
- **[NEW]** = Novel contribution not in any reference
- **[Prof. Kiran]** = Directly from Prof. Kiran's guidance

## Contents

**1 Lineage Table: Every Component Traced to Source** **4**

**2 Study Area: India's Biogeographic Zones [Prof. Kiran]** **6**
2.1 India's 16 Major Forest Types (Champion & Seth Classification) . . . . . 6
2.2 Selected Zones for This Study [Prof. Kiran] . . . . . 6
2.3 Data Volume Estimate [Prof. Kiran] . . . . . 7

**3 Temporal Design: 2003–2025 [Prof. Kiran]** **7**
3.1 Why 22 Years? . . . . . 7
3.2 ENSO and Its Fire Connection [Prof. Kiran] + [NEW] . . . . . 7
3.3 Temporal Split [Prof. Kiran] . . . . . 8

**4 Data Acquisition: MODIS-Consistent Pipeline [Prof. Kiran] + [Doc4]** **8**
4.1 Why MODIS, Not VIIRS [Prof. Kiran] . . . 8
4.2 Complete Data Sources via GEE [Doc4] . . . 8

**5 Preprocessing Pipeline** **8**
5.1 Step 1: Spatial Thinning [Doc1] . . . 8
5.2 Step 2: Non-Fire Point Sampling [Doc2] . . . 9
5.3 Step 3: Multicollinearity Screening [Doc1, Doc2, Doc3] . . . 9
5.4 Step 4: SMOTE on Training Set Only [Doc2] . . . 10
5.5 Step 5: Feature Scaling [Doc2, Doc3] . . . 10

**6 Feature Engineering** **10**
6.1 Base Features (16 Variables, All from Reference Papers) . . . 10
6.2 Cross-Modal Interaction Features [Doc4] . . . 10
6.3 Temporal Lag Features [NEW] + [Prof. Kiran] . . . 10
6.4 Spatial Neighbor Features [NEW] . . . 10
6.5 Total Feature Space . . . 11

**7 Driver Group Weighting (Simplified from [Doc4])** **11**

**8 Multi-Layer Stacking Ensemble [Doc2, Doc3]** **12**
8.1 Layer 1: Five Diverse Base Learners . . . 12
8.1.1 Random Forest [Doc1, Doc2, Doc4] . . . 12
8.1.2 XGBoost [Doc2, Doc3] . . . 12
8.1.3 SVM (RBF) [Doc1, Doc2, Doc3] . . . 12
8.1.4 MLP [Doc3 (BIPE)] . . . 12
8.1.5 LightGBM [Doc4] . . . 13
8.2 Cross-Validated Meta-Features [Doc2, Doc3] . . . 13
8.3 Layer 2: Meta-Learner . . . 13

**9 Temporal Forecasting [NEW]** **13**
9.1 Primary Mode: Lag Features in Ensemble . . . 13
9.2 Optional: Small LSTM for Multi-Month Forecast [NEW] . . . 13

**10 Conformal Prediction & SHAP [NEW] + [Doc2, Doc4]** **14**
10.1 Conformal Prediction [NEW] . . . 14
10.2 Three Uncertainty Sources . . . 14
10.3 SHAP Interpretability [Doc2, Doc4] . . . 14

**11 Per-Zone vs Cross-Zone Modeling [Prof. Kiran]** **14**
11.1 Strategy: Per-Zone Calibration [Prof. Kiran] . . . 14
11.2 Per-Zone Data Size . . . 15

**12 Evaluation Framework** **15**
12.1 Metrics [Doc1–4] + [NEW] . . . 15
12.2 Ablation Study . . . 15
12.3 Statistical Tests [Doc1, Doc2] . . . 15

**13 Computational Considerations [Prof. Kiran]** **16**

**14 Complete Training Pipeline** **17**

**15 Implementation Stack** **17**

**16 v3 → v4 Changes Summary** **18**

**17 Multi-Target Modeling: Location, Extent & Intensity** **18**
17.1 Target 1: Fire Vulnerability (Location) — Primary Task . . . . . 18
17.2 Target 2: Burned Area (Extent) — Secondary Task **[NEW]** . . . . . 18
17.3 Target 3: Fire Intensity (FRP) — Feature & Output **[NEW]** . . . . . 19
17.4 Alignment with Problem Statement . . . . . 19

**18 Novel Contributions** **19**

# 1 Lineage Table: Every Component Traced to Source

| Component | Source | How We Use It |
|---|---|---|
| ***Study Design*** | | |
| Biogeographic zones as units | **[Prof. Kiran]** | 4–5 selected zones (not 16 NPs) |
| Temporal window 2003–2025 | **[Prof. Kiran]** | 22 years of MODIS-consistent data |
| ENSO cycle consideration | **[Prof. Kiran]** | 4–5 yr cycle as temporal features |
| 80/20 cal/val split | **[Prof. Kiran]** | Temporal split (train 2003–2020, test 2021–2025) |
| ***Preprocessing*** | | |
| Spatial thinning | Doc1 | Remove spatial autocorrelation |
| VIF + Pearson screening | Doc1, Doc2, Doc3 | Remove multicollinear features |
| SMOTE class balancing | Doc2 | Handle fire/non-fire imbalance |
| GEE cloud data pipeline | Doc4 | All data from Google Earth Engine |
| ***Feature Engineering*** | | |
| Cross-modal interaction features | Doc4 | terrain×veg, climate×fuel |
| Lag features (3, 6, 12, 24, 60 mo.) | NEW | Temporal context incl. ENSO |
| ENSO index (ONI) as feature | NEW + **[Prof. Kiran]** | Oceanic Niño Index |
| Seasonal encoding | NEW | India fire seasonality |
| Spatial neighbor features | NEW | Moran's $I$, fire spread risk |
| ***Driver Weighting*** | | |
| Learnable softmax group weights | Simplified from Doc4 | 4 weights, 4 driver groups |
| ***Ensemble*** | | |
| Multi-layer stacking | Doc2 | Core ensemble framework |
| Cross-validated meta-features | Doc2, Doc3 | Prevent data leakage |
| RF, XGBoost, SVM, MLP, LightGBM | Doc1–4 | 5 diverse base learners |
| XGBoost / LR meta-learner | Doc2, Doc3 | Options from both papers |
| ***Temporal Forecasting*** | | |
| Lag-feature ensemble (primary) | NEW | Tree models learn temporal splits |
| Small LSTM (optional future mode) | NEW | 3-month-ahead forecast |
| ***Post-Processing*** | | |
| Conformal prediction | NEW | Distribution-free uncertainty |
| SHAP interpretability | Doc2, Doc4 | Feature importance |
| ***Evaluation*** | | |
| AUC-ROC, F1, MCC, Kappa, TSS | Doc1–4 | Comprehensive metrics |

| **Component** | **Source** | **How We Use It** |
|---|---|---|
| Conformal coverage & set size | NEW | Uncertainty metrics |

# 2 Study Area: India's Biogeographic Zones [Prof. Kiran]

## 2.1 India's 16 Major Forest Types (Champion & Seth Classification)

India's forests are classified into 16 major types (Champion & Seth, 1968), whose distribution is coupled with biogeography — climate, altitude, latitude, and rainfall patterns determine which forest type occurs where.

| # | Forest Type | Primary Region | Fire Proneness |
|---|---|---|---|
| 1 | Tropical Wet Evergreen | Western Ghats, NE | Low (wet) |
| 2 | Tropical Semi-Evergreen | Western Ghats, NE | Moderate |
| 3 | Tropical Moist Deciduous | Central, Western India | **High** |
| 4 | Littoral & Swamp | Coasts, Sundarbans | Low |
| 5 | **Tropical Dry Deciduous** | Central India, Deccan | **Very High** |
| 6 | Tropical Thorn | Rajasthan, Gujarat | Moderate |
| 7 | Tropical Dry Evergreen | SE Coast (TN) | Moderate |
| 8 | Sub-Tropical Broad-leaved Hill | NE Hills, Meghalaya | Moderate |
| 9 | **Sub-Tropical Pine** | Himalayas (Chir Pine) | **Very High** |
| 10 | Sub-Tropical Dry Evergreen | Shivalik Hills | High |
| 11 | Montane Wet Temperate | High Himalayas, Nilgiris | Low-Moderate |
| 12 | **Himalayan Moist Temperate** | NW Himalayas | **High** |
| 13 | Himalayan Dry Temperate | Ladakh fringes | Low |
| 14 | Sub-Alpine | High altitude | Low |
| 15 | Moist Alpine Scrub | Very high altitude | Very Low |
| 16 | Dry Alpine Scrub | Essentially treeline | Very Low |

Table 2: India's 16 forest types — bold = most fire-prone, suitable for study

## 2.2 Selected Zones for This Study [Prof. Kiran]

Following Prof. Kiran's directive to **"decrease spatial window"** and focus on **"a few of these zones,"** we select **4–5 biogeographic zones** that are:

1. Most fire-prone (highest annual fire counts in MODIS data)
2. Ecologically diverse (different fire drivers, different climate regimes)
3. Well-covered by MODIS (sufficient data density for 22 years)

| Zone | Forest Type | Representative Region | Why Selected |
|---|---|---|---|
| **Z1** | Tropical Dry Deciduous | Central India (MP, Chhattisgarh, Maharashtra) | Highest fire count in India; **jhum + dry fuel** |
| **Z2** | Sub-Tropical Pine | Uttarakhand, HP (Chir Pine belt) | Pine needle fires; **topography-driven** |
| **Z3** | Tropical Moist Deciduous | Odisha, Jharkhand, Eastern Ghats | High fire + **tribal practice** driven |
| **Z4** | Himalayan Moist Temperate | NW Himalayas (J&K, HP) | **Climate-change sensitive**; increasing fire trend |
| **Z5** | Tropical Semi-Evergreen | NE India (Assam, Mizoram, Nagaland) | **Jhum cultivation** fire; NE hotspot [Doc1] |

Table 3: 5 selected biogeographic zones — diverse fire drivers, manageable spatial scope

**Why these 5? Each zone has a different primary fire driver:**

- Z1: **Dry fuel accumulation** + human ignition
- Z2: **Pine needle litter** + steep slopes + spring dryness
- Z3: **Tribal practice** (Podu/Bewar cultivation) + monsoon failure
- Z4: **Climate change** + westerly disturbance disruption
- Z5: **Jhum (shifting cultivation)** + post-monsoon drying [Doc1]

**The model is calibrated and validated per zone** [Prof. Kiran] — each zone becomes a fundamental unit, and cross-zone transfer can be tested as an additional experiment.

### 2.3 Data Volume Estimate [Prof. Kiran]

| Zone | Approx. MODIS fire points (2003–2025) | After spatial thinnin |
|---|---|---|
| Z1 (Tropical Dry Deciduous) | ∼150,000 | |
| Z2 (Sub-Tropical Pine) | ∼30,000 | |
| Z3 (Tropical Moist Deciduous) | ∼80,000 | |
| Z4 (Himalayan Moist Temperate) | ∼15,000 | |
| Z5 (NE Semi-Evergreen) | ∼60,000 | |
| **Total (5 zones)** | **∼335,000** | ∼ |

Table 4: Estimated data volume — manageable subset of India's ∼500K total [Prof. Kiran]

**After thinning + balanced sampling:** ∼101K fire + 101K non-fire = **∼202,000 samples**. This is **96**× more data than v3's 2,112 samples, eliminating most overfit concerns.

## 3 Temporal Design: 2003–2025 [Prof. Kiran]

### 3.1 Why 22 Years?

The professor's reasoning:

1. **Climate influence requires 2–3 decades** [Prof. Kiran]— short windows miss drought cycles, ENSO effects, and long-term vegetation changes
2. **MODIS consistent from 2003** [Prof. Kiran]— Terra MODIS launched 2000, Aqua 2002, but consistent fire products (Collection 6+) from **2003**
3. **ENSO cycle is 4–5 years** [Prof. Kiran]— with 22 years we capture 4–5 complete ENSO cycles, enough to learn the fire-ENSO relationship
4. 22 years × 12 months = **264 monthly time steps** per spatial unit

### 3.2 ENSO and Its Fire Connection [Prof. Kiran] + [NEW]

**Definition 1** (El Niño–Southern Oscillation (ENSO))**.** *ENSO is a coupled ocean-atmosphere climate phenomenon in the tropical Pacific with a 4–5 year cycle:*

- ***El Niño:*** *Warm Pacific SSTs → reduced Indian monsoon rainfall → drought →* ***more fires***
- ***La Niña:*** *Cool Pacific SSTs → enhanced monsoon → wet conditions → fewer fires*

- ***Neutral:*** *Normal conditions*

The **Oceanic Niño Index (ONI)** is the standard ENSO indicator:

$$\text{ONI}_t = \text{3-month running mean of SST anomaly in Niño 3.4 region (5°N–5°S, 120°–170°W)} \tag{1}$$

- ONI $> +0.5$°C for 5+ consecutive months = El Niño episode
- ONI $< -0.5$°C for 5+ consecutive months = La Niña episode

**How we use ENSO:**

- ONI value as a direct feature (monthly) **[NEW]**
- ENSO phase as categorical: El Niño / La Niña / Neutral **[NEW]**
- ONI lag features: ONI at $t-6$, $t-12$, $t-24$ months (delayed drought effect) **[NEW]**
- IOD (Indian Ocean Dipole) — a secondary mode affecting Indian rainfall **[NEW]**

### 3.3 Temporal Split [Prof. Kiran]

The professor specified **80/20 calibration/validation**:

| Set | Period | Years | % of data |
|---|---|---|---|
| Calibration (Train) | 2003–2020 | 18 years | 80% |
| Validation (Test) | 2021–2025 | 5 years | 20% |
| *Within Calibration, we further split:* | | | |
| Training (inner) | 2003–2018 | 16 years | 72% of total |
| Conformal Calibration | 2019–2020 | 2 years | 8% of total |

Table 5: Temporal split following Prof. Kiran's 80/20 directive

**Why temporal split, not random split:**

- Prevents future information leaking into training (a rain event in 2022 shouldn't help predict 2018)
- Tests true **forecasting ability** — model trained on past, evaluated on future
- The 2021–2025 test period spans ∼1 complete ENSO cycle

## 4 Data Acquisition: MODIS-Consistent Pipeline [Prof. Kiran] + [Doc4]

### 4.1 Why MODIS, Not VIIRS [Prof. Kiran]

### 4.2 Complete Data Sources via GEE [Doc4]

## 5 Preprocessing Pipeline

### 5.1 Step 1: Spatial Thinning [Doc1]

Sarkar et al.'s approach, applied per zone:

$$d(p_i, p_j) \geq d_{\min} = 1 \text{ km} \quad \forall\, p_i, p_j \text{ in same zone} \tag{2}$$

50 bootstrap iterations; keep iteration with maximum retained points [Doc1].

| | **MODIS** | **VIIRS** |
|---|---|---|
| Available from | **2003** | 2012 |
| Temporal coverage for our study | **22 years** | 13 years |
| Spatial resolution | 1 km | 375 m |
| Consistency | **Same sensor 22 yrs** | Shorter record |
| Prof. Kiran's choice | **Yes** | No |

Table 6: MODIS chosen for temporal consistency over VIIRS resolution **[Prof. Kiran]**

| **Dataset** | **GEE Asset ID** | **Resolution** | **Period** | **Paper** |
|---|---|---|---|---|
| Active Fire | MODIS/061/MOD14A1 | 1km | 2003–2025 | **[Prof. Kiran]** |
| Active Fire (FRP) | MODIS/061/MOD14A1 (FRP field) | 1km | 2003–2025 | **[NEW]** |
| Burned Area | MODIS/061/MCD64A1 | 500m | 2003–2025 | Doc4 |
| NDVI/EVI | MODIS/061/MOD13A2 | 1km, 16-day | 2003–2025 | All |
| LAI/FPAR | MODIS/061/MOD15A2H | 500m, 8-day | 2003–2025 | Doc2 |
| Temperature | ECMWF/ERA5_LAND/MONTHLY | 9km | 2003–2025 | Doc2, Doc4 |
| Precipitation | ECMWF/ERA5_LAND/MONTHLY | 9km | 2003–2025 | All |
| Wind (u, v) | ECMWF/ERA5_LAND/MONTHLY | 9km | 2003–2025 | All |
| Humidity | ECMWF/ERA5_LAND/MONTHLY | 9km | 2003–2025 | Doc2 |
| DEM | USGS/SRTMGL1_003 | 30m | Static | All |
| Land Cover | MODIS/061/MCD12Q1 | 500m | Annual | Doc4 |
| Population | WorldPop/GP/100m/pop | 100m | Annual | Doc1–3 |
| Roads | OSM via GEE | Vector | Static | Doc1–3 |
| **ONI (ENSO)** | **NOAA CPC (external)** | **Monthly** | **2003–2025** | **[Prof. Kiran]** + **[NEW]** |
| **IOD** | **NOAA PSL (external)** | **Monthly** | **2003–2025** | **[NEW]** |

Table 7: Data sources — all MODIS-consistent from 2003, GEE pipeline from Doc4

### 5.2 Step 2: Non-Fire Point Sampling [Doc2]

We retain the **natural class imbalance** (fire is rare) rather than artificially balancing:

- Random sampling of non-fire points from forested areas with zero fire history (2003–2025)
- Ratio: 1:3 (fire : non-fire) — reflects real-world rarity **[Doc2]**
- Minimum 5 km from any fire point **[Doc2]**
- Constrained to same forest type as fire points (prevents sampling urban/water)

### 5.3 Step 3: Multicollinearity Screening [Doc1, Doc2, Doc3]

**Step 3A** — Spearman correlation **[Doc2]**: Remove one from pairs with $|\rho_s| > 0.75$

**Step 3B** — Iterative VIF **[Doc1, Doc2, Doc3]**: Remove highest VIF until all $VIF < 10$

**Note:** VIF screening applies to **base features only** (16 variables). Lag features are inherently correlated with their parent variable — this is fine for tree models (RF, XGBoost, LightGBM) which handle collinearity naturally. For **SVM**, we apply PCA to the lag-feature subset to decorrelate **[NEW]**.

Sarkar: 32 → 23; Shahzad: 34 → 16; BIPE: 19 → 16. We expect ∼20 base features → 14–16 after screening.

### 5.4 Step 4: SMOTE on Training Set Only [Doc2]

Since we retain natural imbalance (1:3), SMOTE is needed to prevent majority-class bias:

$$\mathbf{x}_{\text{syn}} = \mathbf{x}_i + \lambda(\mathbf{x}_{nn} - \mathbf{x}_i), \quad \lambda \sim U(0,1) \tag{3}$$

Applied only within the calibration set (2003–2020), **never** on validation (2021–2025). Target ratio after SMOTE: 1:1.5 (slight under-representation to avoid synthetic noise) [Doc2].

### 5.5 Step 5: Feature Scaling [Doc2, Doc3]

StandardScaler fitted on training set: $x'_j = (x_j - \mu_j^{\text{train}})/\sigma_j^{\text{train}}$

## 6 Feature Engineering

### 6.1 Base Features (16 Variables, All from Reference Papers)

| Group | Features | Source | Paper |
|---|---|---|---|
| Weather ($d_w$=5) | Temperature (max, mean) | ERA5 | Doc2, Doc4 |
| | Relative Humidity / VPD | ERA5 | Doc2 |
| | Wind Speed ($\sqrt{u^2+v^2}$) | ERA5 | Doc1, Doc2, Doc4 |
| | Precipitation | ERA5 | All |
| | Drought Index (SPI) | Derived | NEW |
| Fuel ($d_f$=4) | NDVI | MODIS | All |
| | EVI | MODIS | Doc1, Doc4 |
| | LAI | MODIS | Doc2 |
| | Live Fuel Moisture (LFMC) | Derived | NEW |
| Topo ($d_p$=4) | Elevation | SRTM | All |
| | Slope | Derived | All |
| | Aspect | Derived | Doc1–3 |
| | TWI | Derived | Doc1, Doc2 |
| Human ($d_h$=3) | Distance to Roads | OSM | Doc1–3 |
| | Distance to Settlements | WorldPop | Doc2, Doc3 |
| | Population Density | WorldPop | Doc1–3 |

### 6.2 Cross-Modal Interaction Features [Doc4]

Proven top-3 important in FireRisk-Multi:

1. **Terrain-Vegetation:** terrain_veg = Slope $\times$ (1 − NDVI) [Doc4]
2. **Climate-Fuel:** climate_fuel = Temp $\times$ (1 − NDVI) [Doc4]
3. **Risk Contrast:** risk_contrast $= (y_{t-1} - \bar{y}_{\text{neighbors}})/(\text{Std}_{\text{neighbors}} + \epsilon)$ [Doc4]

### 6.3 Temporal Lag Features [NEW] + [Prof. Kiran]

With 22 years, we can now include **multi-year lags** to capture ENSO cycles [Prof. Kiran]:

### 6.4 Spatial Neighbor Features [NEW]

1. **Neighbor Fire Rate:** $\text{NFR} = \frac{1}{|\mathcal{N}|}\sum_{j\in\mathcal{N}} y_{j,t-1}$ [NEW]
2. **Local Moran's $I$:** $I_i = \frac{(y_i-\bar{y})\sum_j w_{ij}(y_j-\bar{y})}{\sum_j w_{ij}}$ [NEW]
3. **Wind-Aligned Fire Spread:** $\text{FS}_i = \max_{j\in\mathcal{N}}[y_{j,t-1}\cdot\cos(\theta_{\text{wind}} - \theta_{j\to i})]$ [NEW]

| **Feature Type** | **Lags** | **Purpose** | **So** |
|---|---|---|---|
| Dynamic variable lags | $t{-}3, t{-}6, t{-}12$ months | Recent + seasonal | [N |
| **Multi-year lags** | $t{-}24, t{-}36, t{-}48, t{-}60$ months | **ENSO cycle** (4–5 yr) | [P |
| Rolling mean (3, 12 months) | $\bar{x}_{t-2:t},\ \bar{x}_{t-11:t}$ | Smoothed trend | [N |
| Rolling std (3, 12 months) | $\sigma_{t-2:t},\ \sigma_{t-11:t}$ | Volatility | [N |
| Fire history | $y_{t-1}, y_{t-12}$ | Recent/annual fire | [N |
| Seasonal encoding | $\sin(2\pi m/12), \cos(2\pi m/12)$ | Month-of-year | [N |
| **ONI (ENSO index)** | Current + $t{-}6, t{-}12$ | El Niño/La Niña state | [P |
| **IOD index** | Current + $t{-}6$ | Indian Ocean influence | [N |
| Cumulative dry days | Count of months with Precip $<$ 20mm in past 6 months | Drought severity | [N |

Table 8: Temporal features — now includes multi-year ENSO lags per Prof. Kiran's directive

### 6.5 Total Feature Space

| **Category** | **Count** |
|---|---|
| Base features (post VIF) | ∼14 |
| Cross-modal features [Doc4] | 3 |
| Temporal lag features **[NEW]** + **[Prof. Kiran]** | ∼45 |
| Climate indices (ONI, IOD) **[Prof. Kiran]** + **[NEW]** | 5 |
| Spatial neighbor features **[NEW]** | 3 |
| Seasonal encoding | 2 |
| **Total** | **∼72 features** |

With ∼200K samples and ∼72 features, the feature-to-sample ratio is ∼2,800:1 — **highly favorable** for all model types including ensembles.

## 7 Driver Group Weighting (Simplified from [Doc4])

Learnable softmax weights for 4 driver groups:

$$[\alpha_w, \alpha_f, \alpha_p, \alpha_h] = \text{softmax}(\mathbf{w}), \quad \mathbf{w} \in \mathbb{R}^4 \tag{4}$$

Weighted feature assembly:

$$\mathbf{x}_{\text{weighted}} = \alpha_w \cdot \tilde{\mathbf{x}}^{\text{weather}} \| \alpha_f \cdot \tilde{\mathbf{x}}^{\text{fuel}} \| \alpha_p \cdot \tilde{\mathbf{x}}^{\text{topo}} \| \alpha_h \cdot \tilde{\mathbf{x}}^{\text{human}} \tag{5}$$

where $\tilde{\mathbf{x}}^{(\cdot)}$ includes base + lag + cross-modal features for each group.

**Per-zone weights [Prof. Kiran]**: Since each zone has different fire drivers, we learn separate $\mathbf{w}_z$ for each zone $z$. This captures that Z2 (Pine) is topography-driven while Z1 (Dry Deciduous) is fuel+weather-driven.

**How are weights learned?** Since tree models are non-differentiable, we cannot backpropagate through the softmax. Two options:

1. **Bayesian Optimization [NEW]**: Treat $\mathbf{w}_z$ as hyperparameters. Use Optuna to optimize $\mathbf{w}_z$ by maximizing validation AUC over 50 trials per zone.

2. **SHAP-Derived [Doc2]**: After initial training with equal weights, compute SHAP group importance → set $\alpha_g \propto \text{mean}|\text{SHAP}_g|$.

We use Option 1 (Bayesian) as primary, Option 2 as sanity check.

**Params:** 4 weights × 5 zones = 20 total. Negligible overfit risk.

# 8 Multi-Layer Stacking Ensemble [Doc2, Doc3]

The core of the methodology — proven by Doc2 (96.2%) and Doc3 (97%, 0.99 AUC).

## 8.1 Layer 1: Five Diverse Base Learners

| Model | Why Selected | Key Hyperparams | Source |
|---|---|---|---|
| Random Forest | Non-linear, robust, OOB error | $n$=500, $m$=$\sqrt{d}$, depth=20 | [Doc1, Doc2, Doc4] |
| XGBoost | Regularized boosting, top in Doc2/3 | lr=0.1, depth=8, $n$=500 | [Doc2, Doc3] |
| SVM (RBF) | Kernel-based boundary | $C$=10, $\gamma$=auto | [Doc1, Doc2, Doc3] |
| MLP | Neural network in ensemble (BIPE) | [256, 128, 64], dropout=0.3 | [Doc3 (BIPE)] |
| LightGBM | Leaf-wise, GOSS, different from XGB | lr=0.05, depth=8, $n$=300 | [Doc4] |

Table 9: Base learners — with 200K samples, MLP can be larger than v3

**Note on MLP size:** With ∼200K samples, MLP [256,128,64] has ∼27K params → ratio = 200K/27K ≈ 7.4 samples/param — **safe**. (In v3 with 1,267 samples we were forced to use [64,32].)

### 8.1.1 Random Forest [Doc1, Doc2, Doc4]

$$\hat{y}_{\mathrm{RF}}(\mathbf{x}) = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{x}; \mathcal{D}_b^*) \tag{6}$$

Split criterion (Gini): $(j^*, s^*) = \arg\max_{j,s}[\mathrm{Gini}(n) - \frac{|n_L|}{|n|}\mathrm{Gini}(n_L) - \frac{|n_R|}{|n|}\mathrm{Gini}(n_R)]$

### 8.1.2 XGBoost [Doc2, Doc3]

$$\mathcal{L}^{(t)} = \sum_i \ell(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \gamma T + \frac{1}{2}\lambda \sum_j w_j^2 \tag{7}$$

Optimal leaf: $w_j^* = -G_j/(H_j + \lambda)$ where $G_j = \sum_{i \in I_j} g_i$, $H_j = \sum_{i \in I_j} h_i$

### 8.1.3 SVM (RBF) [Doc1, Doc2, Doc3]

$$f(\mathbf{x}) = \mathrm{sign}\left(\sum_{i \in \mathrm{SV}} \alpha_i y_i \exp(-\gamma \|\mathbf{x} - \mathbf{x}_i\|^2) + b\right) \tag{8}$$

**Scalability note for SVM:** With 200K samples, standard SVM is slow ($O(n^2$–$O(n^3)$). We use `LinearSVC` or subsample the training set for SVM (50K random subset per fold). Alternatively, replace with SGDClassifier (hinge loss) for scalability.

### 8.1.4 MLP [Doc3 (BIPE)]

Architecture: Input → 256 → 128 → 64 → 1, with BatchNorm + Dropout(0.3) per layer.

Training: Adam optimizer, BCE loss, early stopping patience=15, L2 weight decay $10^{-4}$.

#### 8.1.5 LightGBM [Doc4]

Leaf-wise growth + GOSS + EFB. Config: `n_estimators`=300, `max_depth`=8, `learning_rate`=0.05, `reg_alpha`=0.1, `reg_lambda`=0.5.

### 8.2 Cross-Validated Meta-Features [Doc2, Doc3]

$K = 5$ stratified folds (stratified by zone + class):

$$\hat{p}_i^{(b)} = f_b^{(-\kappa(i))}(\mathbf{x}_{\text{weighted},i}) \quad \Rightarrow \quad \mathbf{M} = [\hat{\mathbf{p}}^{(1)} \| \ldots \| \hat{\mathbf{p}}^{(5)}] \in \mathbb{R}^{N \times 5} \tag{9}$$

### 8.3 Layer 2: Meta-Learner

**Option A:** XGBoost with passthrough [Doc2]: $\hat{y} = g_{\text{XGB}}(\mathbf{M}_i \| \mathbf{x}_i)$

**Option B:** Logistic Regression [Doc3]: $\hat{y} = \sigma(\alpha_1 P_1 + \ldots + \alpha_5 P_5 + b)$

Both tested; select via validation AUC [Doc2, Doc3]. Nested CV for meta-learner to prevent leakage.

## 9 Temporal Forecasting [NEW]

### 9.1 Primary Mode: Lag Features in Ensemble

The temporal lag features (Section 7.3) encode all temporal information directly. Tree models learn splits like:

- "If NDVI dropped > 0.2 over past 3 months AND temperature rose > 5°C → high fire risk"
- "If ONI > 0.5 at $t-6$ AND precipitation below 20th percentile → El Niño drought fire" [Prof. Kiran]

This is temporal reasoning **without a separate temporal model**.

### 9.2 Optional: Small LSTM for Multi-Month Forecast [NEW]

For predicting $\tau = 3$ months ahead ("What will fire risk be in summer?"):

- Input: $W = 12$ months of ensemble predictions + top features
- LSTM: 1 layer, hidden=32 (now safe with 200K samples)
- Zone embedding: $\mathbf{e}_z \in \mathbb{R}^{16}$ (5 zones × 16 dims = 80 params)
- Total params: ∼8,000 — safe for 200K samples

**LSTM equations:**

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f) \tag{10}$$
$$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i) \tag{11}$$
$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c) \tag{12}$$
$$\mathbf{h}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o) \odot \tanh(\mathbf{c}_t) \tag{13}$$

# 10 Conformal Prediction & SHAP [NEW] + [Doc2, Doc4]

## 10.1 Conformal Prediction [NEW]

On calibration set (2019–2020, ∼8% of data):

$$s_i = 1 - \hat{p}_{y_i}, \quad \hat{q} = \text{Quantile}\left(\{s_i\};\ \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right) \tag{14}$$

$$\mathcal{C}(\mathbf{x}_{\text{new}}) = \{c \in \{0,1\} : \hat{p}_c \geq 1 - \hat{q}\} \tag{15}$$

**Proposition 1** (Coverage Guarantee)**.** *$P(Y_{new} \in \mathcal{C}(X_{new})) \geq 1 - \alpha$ for any model and any distribution.*

## 10.2 Three Uncertainty Sources

**1. Conformal Prediction [NEW]**
**2. MC Dropout ($T$=50 passes) [NEW]**: $\text{Var}_{\text{MC}} = \frac{1}{T}\sum_t \hat{p}_t^2 - (\frac{1}{T}\sum_t \hat{p}_t)^2$
**3. Ensemble Disagreement [NEW]**: $\text{Var}_{\text{ens}} = \frac{1}{5}\sum_{b=1}^{5}(\hat{p}^{(b)} - \bar{p})^2$

## 10.3 SHAP Interpretability [Doc2, Doc4]

TreeSHAP for tree models; four-level analysis:

1. **Group-level:** Which driver group? Cross-check with $\alpha_w, \alpha_f, \alpha_p, \alpha_h$ per zone
2. **Feature-level:** Which specific variable? [Doc2]
3. **Temporal-level:** Which lag period (ONI-lag-12 vs temp-lag-3)? [NEW]
4. **Zone-level:** How do top features differ between Z1 vs Z2 vs Z3? [Prof. Kiran]

# 11 Per-Zone vs Cross-Zone Modeling [Prof. Kiran]

## 11.1 Strategy: Per-Zone Calibration [Prof. Kiran]

The professor stated zones are "fundamental units for model cal/val." So:

**Primary approach:** Train a **separate ensemble per zone**. Each zone's model learns its own fire drivers.

- Z1 model: learns dry fuel + temperature dominance
- Z2 model: learns pine litter + slope + spring drying
- Z3 model: learns tribal practice + monsoon failure
- Z4 model: learns climate trend + westerly disturbance
- Z5 model: learns jhum + post-monsoon drying

**Secondary experiment (ablation):** Pool all 5 zones with zone indicator features → test if cross-zone transfer helps or hurts.

| Zone | Samples (est.) | MLP safe? | XGBoost depth OK? |
|---|---|---|---|
| Z1 (Dry Deciduous) | ∼80,000 | Yes, [256,128,64] (27K p, 2.4:1) | Yes, depth=8 |
| Z2 (Pine) | ∼20,000 | Yes, [128,64] (10K p, 1.6:1) | Yes, depth=6 |
| Z3 (Moist Deciduous) | ∼50,000 | Yes, [256,128,64] (27K p, 1.5:1) | Yes, depth=8 |
| Z4 (Himalayan Temperate) | ∼12,000 | **[32,16]** (2.9K p, 3.3:1) | Yes, depth=5 |
| Z5 (NE Semi-Evergreen) | ∼40,000 | Yes, [128,64] (10K p, 3.2:1) | Yes, depth=8 |

Table 10: Per-zone MLP sizing — adapted to each zone's data volume

### 11.2 Per-Zone Data Size

## 12 Evaluation Framework

### 12.1 Metrics [Doc1–4] + [NEW]

| Metric | Formula | Source |
|---|---|---|
| AUC-ROC | Area under ROC | Doc1–4 |
| AUC-PR | Area under PR curve | Doc2 |
| F1-Score | $2PR/(P+R)$ | Doc2 |
| MCC | $(TP{\cdot}TN - FP{\cdot}FN)/\sqrt{\cdots}$ | Doc3 |
| Cohen's Kappa | $(p_o - p_e)/(1 - p_e)$ | Doc3 |
| TSS | Sensitivity + Specificity $-1$ | Doc1 |
| Brier Score | $\frac{1}{N}\sum(y_i - \hat{p}_i)^2$ | Doc2 |
| Conformal Coverage | $P(y \in \mathcal{C})$ | NEW |
| Conformal Set Size | $\frac{1}{N}\sum\lvert\mathcal{C}\rvert$ | NEW |

### 12.2 Ablation Study

| Variant | What Changes | Tests |
|---|---|---|
| AGFE-Fire v4 (Full) | Nothing | Full model per zone |
| −Cross-modal | Remove Doc4 features | Doc4 feature value |
| −Lag features | Remove temporal lags | Temporal contribution |
| −ENSO features | Remove ONI/IOD | Prof's ENSO hypothesis |
| −Multi-year lags | Keep only 3,6,12 lags | ENSO cycle value |
| −Spatial features | Remove Moran's $I$ etc | Spatial contribution |
| −Driver weights | Equal 0.25 each | Weighting value |
| Simple Average | Avg base learners | Stacking value |
| AUC-Weighted [Doc1] | Sarkar's method | Baseline 1 |
| Shahzad Stack [Doc2] | RF+SVM+LR+KNN | Baseline 2 |
| BIPE [Doc3] | SVM+MLP+XGB → LR | Baseline 3 |
| Pooled All-Zone | Single model, 5 zones | Per-zone vs pooled |

Table 11: Ablation includes ENSO-specific test per Prof. Kiran's climate emphasis

### 12.3 Statistical Tests [Doc1, Doc2]

McNemar's test, DeLong's test for AUC, Friedman + Nemenyi post-hoc.

## 13 Computational Considerations [Prof. Kiran]

The professor flagged that "this dataset might go into half a million fire points" and is "computationally intensive." Our approach:

| **Bottleneck** | **Model** | **Solution** |
|---|---|---|
| GEE data download | All | Export as TFRecord/CSV via GEE Tasks (batch, not interactive) [Doc4] |
| SVM training ($O(n^3)$) | SVM | Use SGDClassifier (hinge loss) or subsample 50K per fold |
| RF training | RF | Parallelize: `n_jobs=-1` across CPU cores |
| XGBoost/LightGBM | XGB, LGB | GPU training: `tree_method='gpu_hist'` |
| MLP training | MLP | GPU via PyTorch, batch size 1024 |
| SHAP computation | TreeSHAP | TreeSHAP is $O(TLD)$ — fast for trees; subsample for SHAP summary plots |
| Memory | All | Process per zone (max ∼80K samples per zone fits in 16GB RAM) |

Table 12: Scalability solutions for Prof. Kiran's data volume concern

**Estimated training time per zone:**

- RF (500 trees): ∼2–5 min
- XGBoost (500 rounds): ∼1–3 min (GPU)
- LightGBM: ∼1–2 min
- SVM (SGD, 50K subsample): ∼1 min
- MLP (100 epochs, batch 1024): ∼5 min (GPU)
- 5-fold stacking total: ∼50–75 min per zone
- 5 zones: ∼4–6 hours total

## 14 Complete Training Pipeline

**Algorithm 1** AGFE-Fire v4 — Final Pipeline (Prof. Kiran's Design)

**Require:** MODIS fire data + covariates for 5 biogeographic zones, 2003–2025
**Ensure:** Per-zone fire vulnerability models with uncertainty

```
// Phase 0: Data Acquisition [Doc4 + Prof]
Download MODIS fire, NDVI, LAI, ERA5, SRTM via GEE [Doc4]
Download ONI, IOD indices from NOAA [Prof + NEW]
Aggregate to zone-level pixel grid, monthly composites

// For each zone z ∈ {Z1, Z2, Z3, Z4, Z5}: [Prof]
for zone z = 1 to 5 do

    // Phase 1: Preprocessing [Doc1, Doc2, Doc3]
    Spatial thinning: 1km minimum distance [Doc1]
    Balance: equal non-fire points [Doc2]
    Spearman + VIF screening [Doc1, Doc2, Doc3]
    Temporal split: Train (2003–2018), Cal (2019–2020), Test (2021–2025) [Prof: 80/20]
    SMOTE on training only [Doc2]

    // Phase 2: Feature Engineering [Doc4 + NEW + Prof]
    Cross-modal features [Doc4]: terrain_veg, climate_fuel, risk_contrast
    Lag features [NEW]: 3, 6, 12, 24, 36, 48, 60 month lags
    ENSO features [Prof + NEW]: ONI (current + lags), IOD
    Spatial features [NEW]: neighbor fire rate, Moran's I, fire spread
    Seasonal encoding [NEW]

    // Phase 3: Learn Zone-Specific Driver Weights [simplified Doc4]
    [α_w^(z), α_f^(z), α_p^(z), α_h^(z)] = softmax(w_z)

    // Phase 4: Stacking Ensemble [Doc2, Doc3]
    for fold k = 1 to 5 do
        for base b ∈ {RF, XGB, SVM, MLP, LightGBM} do
            Train f_b^(−k), generate OOF predictions [Doc2, Doc3]
        end for
    end for
    Train meta-learner (XGBoost [Doc2] or LR [Doc3])

    // Phase 5: Calibrate & Explain [NEW + Doc2]
    Conformal calibration on 2019–2020 [NEW]
    SHAP analysis per zone [Doc2, Doc4]
end for

// Phase 6: Cross-Zone Comparison [Prof]
Compare driver importance across zones via SHAP [Doc2]
Test pooled vs per-zone model [Prof: ablation]
```

## 15 Implementation Stack

| Component | Library | Source |
|---|---|---|
| Data acquisition | GEE Python API (`ee`) | Doc4 |
| Core ML | scikit-learn 1.4+ | Doc1–3 |
| XGBoost | XGBoost 2.0+ (GPU) | Doc2, Doc3 |
| LightGBM | LightGBM 4.0+ (GPU) | Doc4 |
| MLP + LSTM | PyTorch 2.1+ (GPU) | Doc3 + NEW |
| Conformal | MAPIE | NEW |
| SHAP | SHAP 0.44+ | Doc2, Doc4 |
| ENSO data | `noaa-coaps` or manual CSV | Prof + NEW |
| Geospatial | geopandas, rasterio | Doc1, Doc4 |
| Visualization | matplotlib, seaborn, folium | All |

# 16 v3 → v4 Changes Summary

| Aspect | v3 | v4 (Prof. Kiran) |
|---|---|---|
| Study area | 16 National Parks | **5 biogeographic zones** [Prof] |
| Temporal | 2014–2024 (11 yr) | **2003–2025 (22 yr)** [Prof] |
| Fire data | VIIRS (from 2012) | **MODIS (from 2003)** [Prof] |
| Data size | ∼2,112 samples | **∼202,000 samples** [Prof] |
| Cal/Val split | 60/15/10/15 | **80/20 temporal** [Prof] |
| ENSO | Not considered | **ONI + IOD as features** [Prof] |
| Multi-year lags | Max 12 months | **Up to 60 months (ENSO cycle)** [Prof] |
| Per-zone models | Single global model | **Separate per zone** [Prof] |
| Modeling scope | All India (shallow) | **5 zones (deep)** [Prof] |
| MLP size | [64, 32] (overfit-safe) | [256, 128, 64] (safe with 200K) |
| SVM scalability | Standard SVC | SGDClassifier / subsample |
| Everything else | Same | Same — stacking, SHAP, conformal all kept |

Table 13: All changes driven by Prof. Kiran's guidance

# 17 Multi-Target Modeling: Location, Extent & Intensity

The problem statement specifies three fire characters: **location, extent, and intensity**. Our approach addresses all three:

## 17.1 Target 1: Fire Vulnerability (Location) — Primary Task

Binary classification: fire ($y = 1$) vs non-fire ($y = 0$) at each pixel. This is the core stacking ensemble described in Section 8.

**Output:** $\hat{p}(\text{fire}|\mathbf{x}) \in [0, 1]$ per pixel per month.

## 17.2 Target 2: Burned Area (Extent) — Secondary Task [NEW]

Using MODIS MCD64A1 burned area product:

- **Target:** $y_{\text{BA}} = \log(1 + \text{burned area in hectares})$ per pixel per month

- **Model:** Same stacking framework but with **regression** (RMSE loss instead of BCE)
- **Meta-learner:** XGBoost regressor [Doc2] or Ridge regression
- Only trained on pixels where fire occurred — avoids zero-inflation

### 17.3 Target 3: Fire Intensity (FRP) — Feature & Output [NEW]

MODIS provides **Fire Radiative Power (FRP)** in MW, measuring fire intensity:

- **As input feature:** Historical mean FRP per pixel (from 2003–$t-1$) serves as a "how intense do fires get here?" feature, added to the base feature set
- **As output (optional):** Regression to predict FRP class (Low $<$ 50MW, Medium 50–200MW, High $>$ 200MW) using ordinal classification

### 17.4 Alignment with Problem Statement

| Fire Character | How Modeled | Data Source |
|---|---|---|
| **Location** | Binary classification (primary) | MOD14A1 active fire |
| **Extent** | Burned area regression (secondary) | MCD64A1 burned area |
| **Intensity** | FRP as feature + optional target | MOD14A1 FRP field |

**Driver categorization per problem statement:**

- **Dynamic drivers (coupled):** Weather (temp, precip, humidity, wind, ENSO) $\times$ Fuel (NDVI, EVI, LAI, LFMC) — captured via cross-modal features [Doc4]
- **Static drivers:** Topography (DEM, slope, aspect, TWI) + Human (roads, settlements, population) — separate groups with learnable weights

## 18 Novel Contributions

**C1. Biogeographic zone-level fire modeling for India [Prof. Kiran] + [NEW]**: First ensemble ML study using India's 16-type forest classification as fundamental units, with per-zone calibration capturing region-specific fire drivers.

**C2. ENSO-aware fire prediction [Prof. Kiran] + [NEW]**: Explicit inclusion of ONI and IOD indices with multi-year lags (up to 60 months) to capture the El Niño–drought–fire cascade. No reference paper includes ENSO.

**C3. 22-year MODIS-consistent temporal analysis:** Longest consistent fire modeling period for India (2003–2025), capturing 4–5 complete ENSO cycles per Prof. Kiran's climate directive.

**C4. Fusion of 4 reference papers:** Stacking (Doc2) + cross-modal features (Doc4) + MLP in ensemble (Doc3) + India-focused feature engineering (Doc1) + VIF screening (Doc1–3) + SHAP (Doc2, Doc4).

**C5. Conformal prediction for fire risk [NEW]**: First distribution-free uncertainty quantification in forest fire ML. Guaranteed coverage.

**C6. Multi-target framework [NEW]**: Predicts fire **location** (classification), **extent** (burned area regression), and **intensity** (FRP) — addressing all three fire characters from the problem statement.

**C7. Overfit-conscious architecture:** Model complexity scaled to per-zone data volume — MLP size, tree depth, and LSTM hidden size all justified by sample count (see Z4 constraint in Table 8).

**Bottom line:** AGFE-Fire v4 follows Prof. Kiran's guidance exactly —

**Temporal:** 2003–2025, 22 years, MODIS-consistent, ENSO-aware
**Spatial:** 5 biogeographic zones (not all India), per-zone calibration
**Split:** 80/20 temporal (no future leakage)
**Targets:** Fire location (classification) + extent (regression) + intensity (FRP)
**Methods:** Best of Doc1 + Doc2 + Doc3 + Doc4, all tagged
**Data:** ∼200K+ samples — MLP/tree sizes tuned per zone to prevent overfit
**Guard-rails:** VIF, PCA for SVM, adaptive MLP sizing, SMOTE only on train

*"Increase temporal window, decrease spatial window"* — Prof. Kiran